

R. Elvert · N. Kronfeld · T. Dayan · A. Haim
N. Zisapel · G. Heldmaier

Telemetric field studies of body temperature and activity rhythms of *Acomys russatus* and *A. cahirinus* in the Judean Desert of Israel

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Abstract Two species of the genus *Acomys* coexist in arid zones of southern Israel. *Acomys russatus* is distributed in extremely arid areas, while *A. cahirinus* is common in both Mediterranean and arid regions. Individuals of both species from a rodent community in the Ein Gedi Nature Reserve were implanted with temperature-sensitive transmitters. Body temperature (T_b) rhythms were recorded in free-ranging mice at four different seasons of the year. *A. cahirinus* (30–45 g) showed a nocturnal rhythm of T_b throughout the year. In the activity phase during the night T_b increased to 38.2°C. During the day T_b decreased to 34°C. This species displayed this pattern in summer also when ambient temperatures rose above T_b . The T_b of *A. russatus* (45–65 g) varied between 34.8 and 41°C during the hot season, showing a bimodal temperature rhythm with maximal values in the morning and in the evening. Measurements of activity in this species showed inactivity during the hottest period of a summer day. In winter *A. russatus* showed no clearly detectable diurnal or ultradian rhythm in T_b , which remained constant between narrow limits of 35.2 and 36.8°C.

Key words *Acomys* · Telemetry · Body temperature · Activity · Circadian rhythms

Introduction

Two species of spiny mice, *Acomys russatus* and *A. cahirinus*, coexist in arid areas in the south of Israel (Shkolnik 1971; Kronfeld et al. 1996). While *A. cahirinus* is widely distributed in the semi-arid and even in the mesic regions of Israel, *A. russatus* is limited to the extremely hot environments of the southern and the central parts of the Rift Valley in Israel (Harrison and Bates 1991) and overlaps with *A. cahirinus* throughout its range. The habitat of the spiny mice in the Judean Desert is characterised by steep rocky mountain escarpments running parallel to the Dead Sea. Huge blocks alternate with small boulders. Spiny mice do not construct burrows, but seek the shelter of rock crevices.

Trapping data suggest that the common spiny mouse (*A. cahirinus*) is active during the night, as are most desert rodents, while the golden spiny mouse (*A. russatus*) is active during the day (Shkolnik 1971). *A. russatus* switched to nocturnal activity when the nocturnal conspecific was removed from the habitat. This implies that interspecific competition could be the driving force for the temporal segregation of activity among the two *Acomys* species. The golden spiny mouse appears to be forced into diurnal activity by its more aggressive congener (Shkolnik 1971). This is supported by laboratory studies demonstrating that *A. russatus* displays nocturnal rhythms of body temperature and metabolic rate when separated from *A. cahirinus* (Haim and Zisapel 1992, 1994; Rubal et al. 1992; Fluxman and Haim 1993; Kronfeld et al. 1994). Chemical cues released by *A. cahirinus* into the habitat may maintain the time separation of activity patterns of these two species (Haim and Rozenfeld 1993; Friedman et al. 1997).

Trapping data do not allow continuous recording of 24-h rhythms in individual mice and thus detailed activity time budgets of undisturbed animals cannot be studied. We applied the technology of body temperature telemetry to individuals of both species where they coexist at Ein Gedi. This technology enables us to study

R. Elvert (✉) · G. Heldmaier
Department of Biology, Philipps-University,
D-35032 Marburg, Germany
e-mail: elvert@mail.uni-marburg.de, Fax: +49-6421-288937

N. Kronfeld · T. Dayan
Department of Zoology
Tel Aviv University,
Tel Aviv 69978, Israel

A. Haim
Department of Biology,
University of Haifa-Oranim,
Kiryat Tiv'on 36910, Israel

N. Zisapel
Department of Biochemistry,
Tel Aviv University,
Tel Aviv 69978, Israel

the (body temperature) T_b rhythm of individual mice, shedding light on their overt temperature rhythms. The transmitters used were further developed in order to record also activity patterns of individual animals, enabling us to study the relationship between actual activity and overt temperature rhythms of free-ranging populations of spiny mice.

Materials and methods

Individuals of *A. cahirinus* and *A. russatus* were trapped on a rocky hill slope of about 4000 m² near the Ein Gedi Field School (31°28'N, 35°23'E, 100–350 m below sea level) at all seasons of the year. The time schedule included three seasons in 1995: February/March, July/August and October/November; and two in 1996: January and October/November. The fall season was repeated because of technical problems and loss of data caused by environmental influences, e.g. flood of the study area in November 1995. We trapped the mice with 50 collapsible Sherman box traps spread randomly in the area. Peanut butter and a small piece of carrot were used as bait. In each study season, traps were set for 25–31 days and they were checked daily just after sunrise and before sunset. In the summer, traps were closed during daytime hours and additionally covered with styrofoam for heat protection. The spiny mice were individually marked with toe clipping. Every individual was weighed whenever trapped using Pesola dynamometer scales with 0.3% accuracy. Population densities and abundances were calculated using the Jolly-Seber method (Krebs 1989; Sutherland 1996) which is designed for open populations where all individuals in the first sample are unmarked. This method requires a previous and following trapping period, so it was only possible to estimate population size for two seasons. The densities can be estimated from relationship of “size of marked population” to “number of marked animals” (Krebs 1989). Because of a delay of 9 months between the trapping period in January 1996 and October/November 1996, the latter study season was not included for population estimates.

Mice which were trapped repeatedly were implanted with temperature transmitters into the abdominal cavity (Oekokart E&V, Munich, Germany). For the implantation, mice were anaesthetised with a mixture of ketamine and xylazine. Both species required different dosages of the anaesthetic (*A. cahirinus*: ketamine 80–110 mg kg⁻¹ and xylazine 4 mg kg⁻¹; *A. russatus*: ketamine 120–150 mg kg⁻¹ and xylazine 5 mg kg⁻¹). Anaesthetics were injected i.p. and transmitters were implanted in the abdominal cavity. The peritoneum was sutured with resorbable catgut (1.5 metric), and the skin with sewing silk (1.5 metric). After a recovery period of 36 h the animals were released again in their natural habitat. During the measuring period, the implanted mice were retrapped repeatedly and never showed infections. At the end of each recording period, the mice were retrapped to remove the transmitters. The transmitters were of FM type with an integrated internal loop-antenna. They transmitted signals in the frequency range of 150–151 MHz. A 3-V lithium battery supplied the transmitters with energy for 25–28 days. Prior to implantation, each transmitter was sealed against moisture with a mixture of paraffin and purified beeswax. The final size of a transmitter was about 17 × 17 × 9 mm and it weighed, with wax coating, 4.5–4.8 g. The transmitters sent 10-ms signals at intervals of 300–1000 ms, and these intervals correlated with T_b . The temperature was calculated from each signal interval following calibration of the transmitters, using the equation: T (in °C) = $A + B \log(t)$ (see Ruf and Heldmaier 1987).

The receiver unit, data logger and computer for control of the receiver and data storage were mounted in an aluminium box. The entire system was powered by a 12-V, 24-Ah battery which was recharged by solar panels (Siemens). The solar panels also provided energy for a peltier cooling system for the receiver setup which was necessary during the hot season when T_{air} s rose above 40°C. Transmitter signals were received through two omnidirectional

antennae (type Ringo Ranger) placed 30 m apart in the study area. Scanning the two antennae improved signal monitoring. Both antennae received the signals with different strength, and this varied with the position of the mice. Changes in signal field strength between two successive records were recorded as activity in October 1996. Constant signal field strength was interpreted as no activity. The tracking receiver (Reichenbach, Freiburg, Germany) was connected via an interface (DACpad-71 B, Datalog. Moenchengladbach, Germany) to a low-energy-consuming notebook-computer (COMPAQ Contura Aero 4/25) which stored the data on hard disk, and controlled the receiver channels as well as antenna scanning. The entire system was controlled by self-developed software (QB45) which also included filter algorithms for suppression of noise and interference.

Ambient temperatures were measured by thermocouples (copper/constantan) placed in the sun (air temperature, 30 cm above ground) and 1 m inside a rock crevice. Temperatures were recorded from the same positions at all seasons. T_b , T_{air} and activity were recorded at 4-min intervals and stored in the computer and a data logger, THERM 2281-8 (AMR, Holzkirchen, Germany).

For statistical analysis, data are expressed as the mean ± SD. Student's *t*-test or Mann-Whitney *U*-test were used for comparisons between mean hourly values of photo- and scotophase.

Results

A. russatus

Twenty-eight individuals of *A. russatus* were trapped on 377 occasions in March, July and October 1995 and in January 1996. All were marked when trapped the first time. Two individuals could be retrapped in each of the first four seasons (Fig. 1). Mouse no. 5000 was retrapped 7 times in March and July 1995, 8 times in October 1995 and 5 times in January 1996. Another individual (no. 0200) which was not implanted with a transmitter was retrapped 19 times in March, 11 times in July, 16 and 17 times in October and January, respectively. In other individuals, the retrapping incidence varied between 5 and 13 times during one season. The density of *A. russatus* in our study site, based on the population estimation using the Jolly-Seber method, was about 38 individuals per hectare during July and 35 individuals per hectare during the October season.

High retrapping rates and continuous monitoring of body weight during different seasons indicate that the sedentary behaviour of mice was not disturbed by our trapping activities and transmitter implantation (Fig. 1). One individual *A. russatus* (no. 5000) which was retrapped with a high frequency had a maximum body weight during the hot season (61.9 and 59.4 g, mean values in July and October, respectively) and lower body weight in the cold season (48.6 g in March and 52.8 g in January). Another individual (no. 0200) served as a control animal and was not implanted with transmitters. It was slightly bigger, but showed similar seasonal changes in body weight with highest values in July and October (66.8 and 68.3 g) and minimal values in January and March (63.9 and 60.9 g). The exceptional high body weight of this individual and its rapid decrease in early March was probably due to giving birth, as concluded from vaginal bleeding detected during early March.

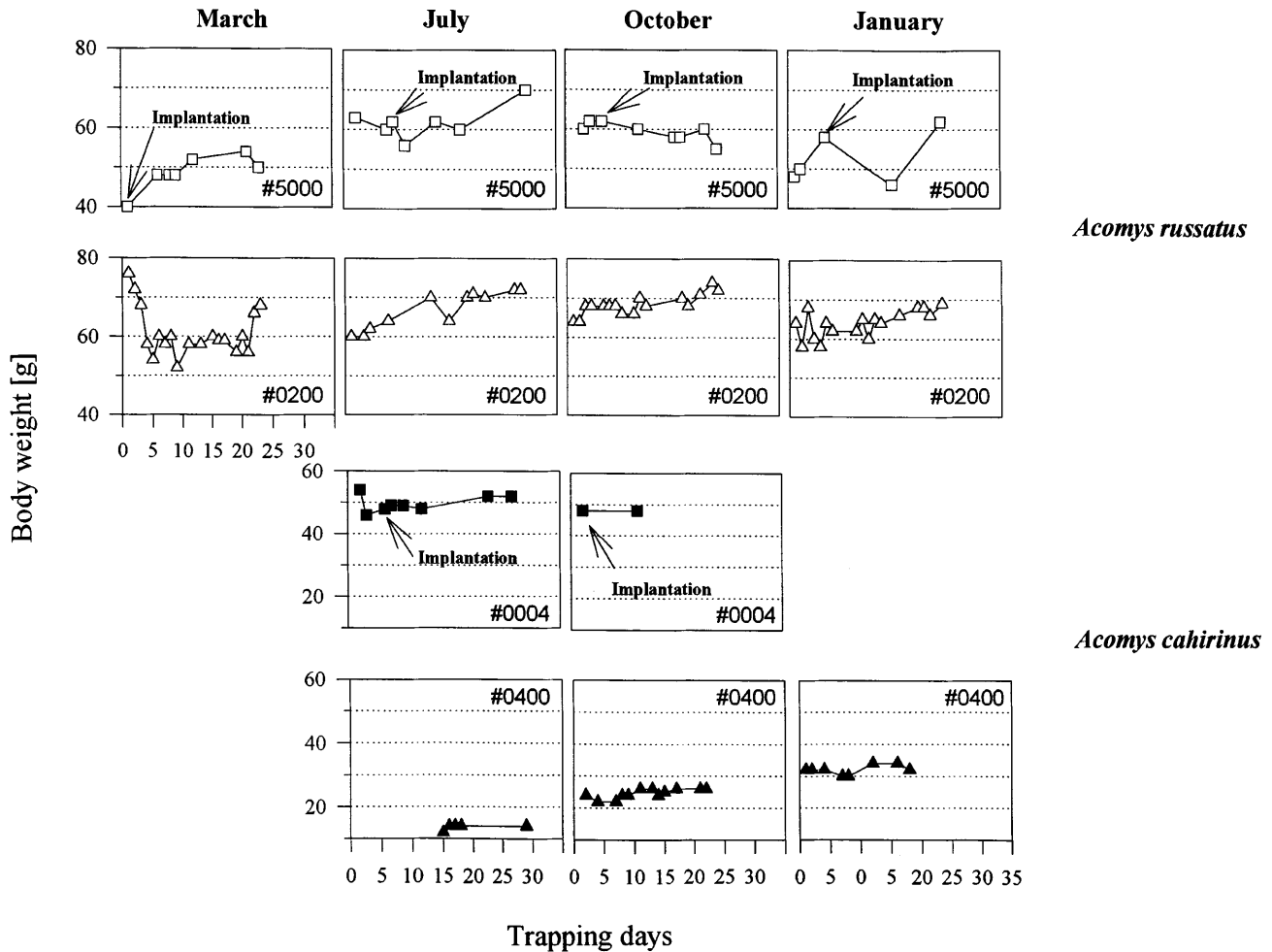


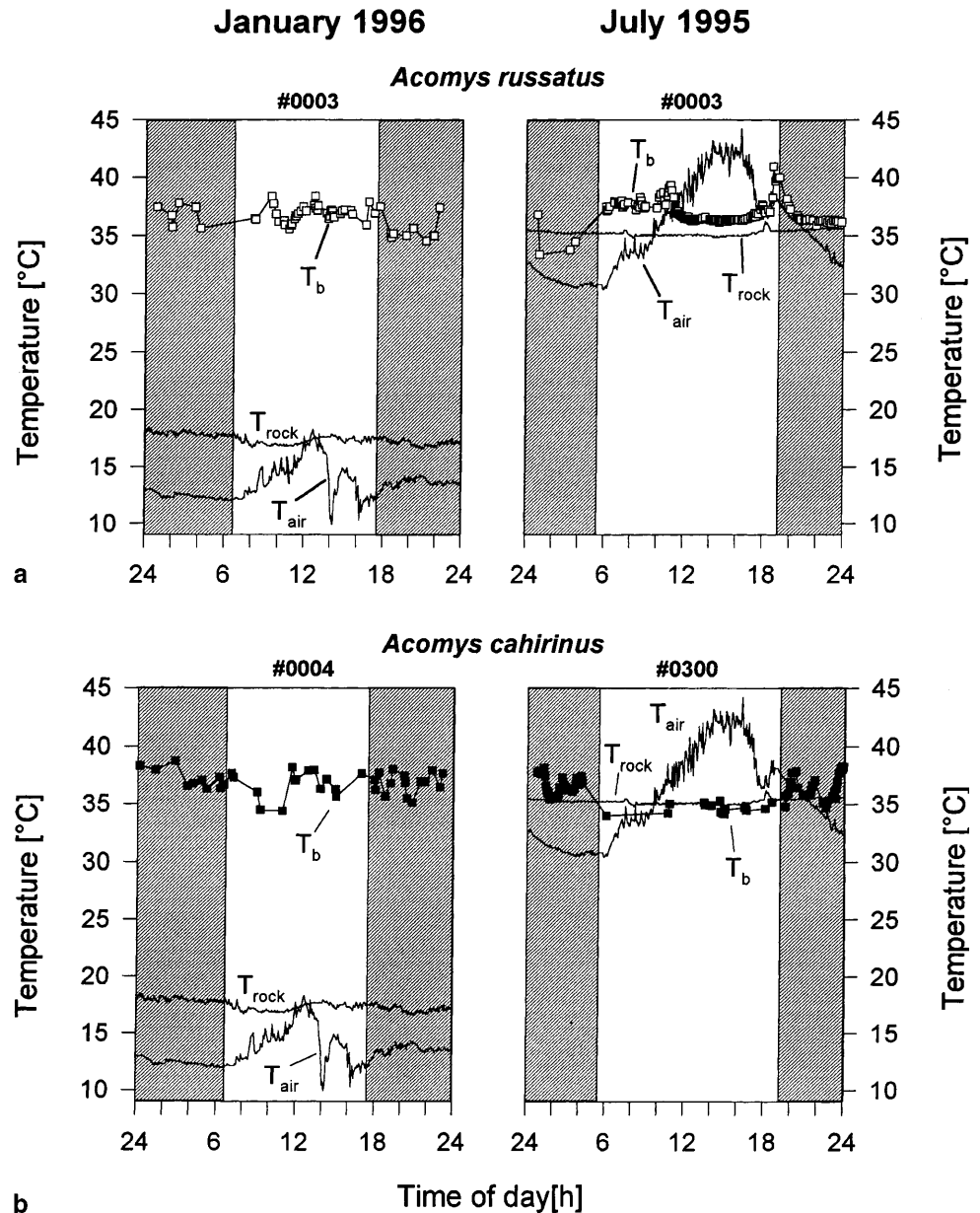
Fig. 1 Body weights of an implanted (no. 5000) and non-implanted (control, no. 0200) *Acomys russatus* over four seasons. Mean $\pm$ SD body weights of individuals in each season: mouse with transmitter, March 48.6 ± 4.1 g, July 61.9 ± 3.9 g, October 59.4 ± 2.2 g, January 52.8 ± 6.1 g; control mouse, March 60.9 ± 6.1 g, July 66.8 ± 4.6 g, October 68.3 ± 2.6 g, January: 63.9 ± 3.4 g. Body weights of an implanted (no. 0004, two seasons) and non-implanted (no. 0400, three seasons) *Acomys cahirinus*. Mean $\pm$ SD body weights: mouse with transmitter, July: 49.8 ± 2.5 g, October 48.0 ± 0 g; mouse without transmitter, July 13.6 ± 0.8 g, October 24.6 ± 1.4 g, January 32.0 ± 1.4 g

Monitoring of body weights showed that seasonal changes were not affected by transmitter implantation. Mean body weight of all trapped individuals was 48.2 g ($n = 28$).

Ambient temperatures showed pronounced seasonal variations in our study area (Fig. 2). In July, T_{air} increased close to 45°C whereas in January minimal temperatures at about 10°C were observed, following short episodes of rainfall. In each season, a distinct day/night variation in T_{air} was observed with minima just before sunrise and maximum values during early afternoon. Despite these daily variations of T_{air} , the temperatures in the sample rock crevice did not show daily variations but remained constant throughout the measuring periods, at about 35°C in summer and 18°C in winter (Fig. 2).

A. russatus (no. 0003, Fig. 2) exhibited higher temperatures during the day than during the night. Mean values per hour on these sample days showed significantly higher T_b values during the day in January and in July ($P < 0.05$, Student's t -test). Highest levels of T_b with values at 41°C were recorded during late morning hours, 4.5 h after sunrise, and during late afternoon, 1.5 h before sunset throughout the hot season, indicating a bimodal pattern of body temperature. During midday hours, T_b remained rather constant at 36.4°C . During the night, T_b was at a lower level (34.8 – 35.7°C) than during the diurnal minimum (Fig. 2). The daily amplitude of T_b variations was 6.2°C . A bimodal pattern of T_b was also observed for an individual during October 1996 (no. 0003) with T_{air} maxima at about 38°C (filled triangles in Fig. 3). An example for a consecutive record of T_b of this golden spiny mouse showed a bimodal pattern in T_b rhythm on 7 out of 11 days (1, 2, 4, 6, 7, 8 and 10). Day 3 and 11 were rather arrhythmic whereas day 5 and 9 indicated a diurnal but not a bimodal T_b rhythm. Statistical testing showed significantly higher T_b values during photophase than during scotophase ($P < 0.05$, Student's t -test). The comparison of mean values per hour of photo- or scotophase of all days together showed no significant difference (Mann-

Fig. 2 Sample 24-h recordings of ambient temperatures and body temperature of *Acomys* individuals during the hot (31 July) and cold season (18 January *A. cahirinus*; 14 January *A. russatus*) in the Judean Desert (T_{rock} temperature inside a rock crevice, T_{air} air temperature in the sun, T_b body temperature, grey area scotophase). **a** *A. russatus*, no. 0003. **b** *A. cahirinus*, no. 0004 in January and no. 0300 in July



Whitney *U*-test and Student's *t*-test). In four mice (nos. 0055, 0013, 0505 and 5000), we also recorded locomotor activity from changes in signal level of the T_b transmitter (no. 0003, Fig. 4). These activity records coincided with monitored T_b patterns, and showed a bimodal pattern of activity (day 8, Fig. 3).

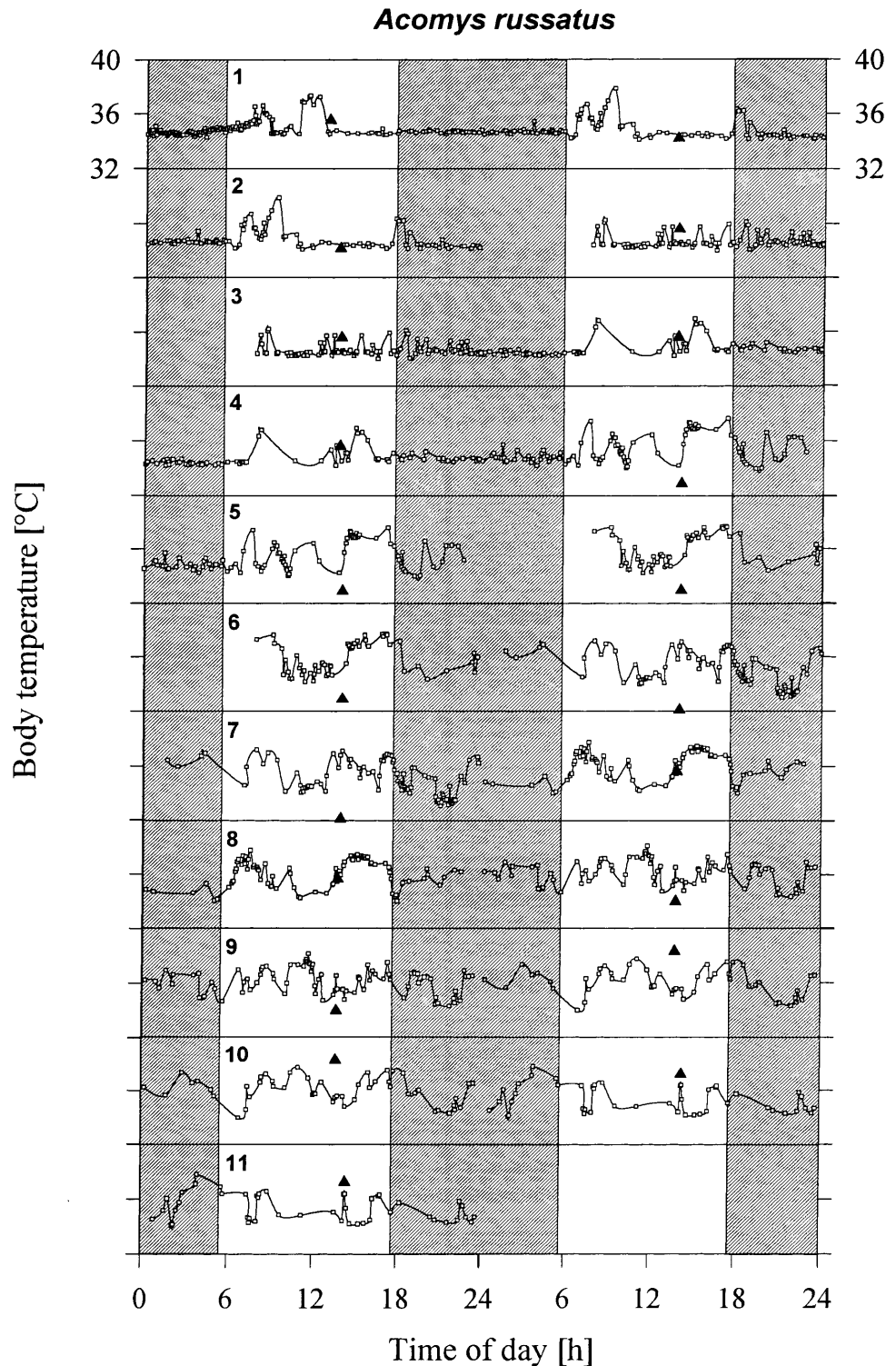
In October, high variability in T_b patterns was observed (Fig. 5a). Whereas two mice showed bimodal T_b patterns (no. 0003 and no. 0013), others exhibited no distinct rhythm (no. 0055, no. 0505 and no. 5000). In January, no significant differences could be detected between photo- and scotophase (36.25°C during day, 36.08°C during night; see mean values of day/night period in Table 1). Individual recordings of T_b received from two implanted mice in January (no. 5000 and no. 0003) varied between 35.2 and 36.8°C during the day and 35.0 and 36.7°C at night (Fig. 5a). T_b values in

March ($n = 2$), July ($n = 2$) and in some individuals in October ($n = 5$) showed significantly higher values during daytime (Fig. 5a).

A. cahirinus

Twenty-six individuals of *A. cahirinus* were trapped a total of 187 times. One mouse (no. 0400) was trapped 25 times during three consecutive seasons. Because of its low body mass (13.6 g in July, 24.6 g in October, 32.0 g in January; Fig. 1) it was not considered for implantation with temperature transmitters. One individual (no. 0004), which was suitable for implantation, could be retrapped in two consecutive seasons (July 9 times and October 2 times). In other individuals of this species, trapping incidences varied between 2 and 16 times. The

Fig. 3 Double plot of continuous body temperature records of *A. russatus* (mouse no. 0003) in the Judean Desert (9–19 October 1996 (closed triangles maximum daily T_a))

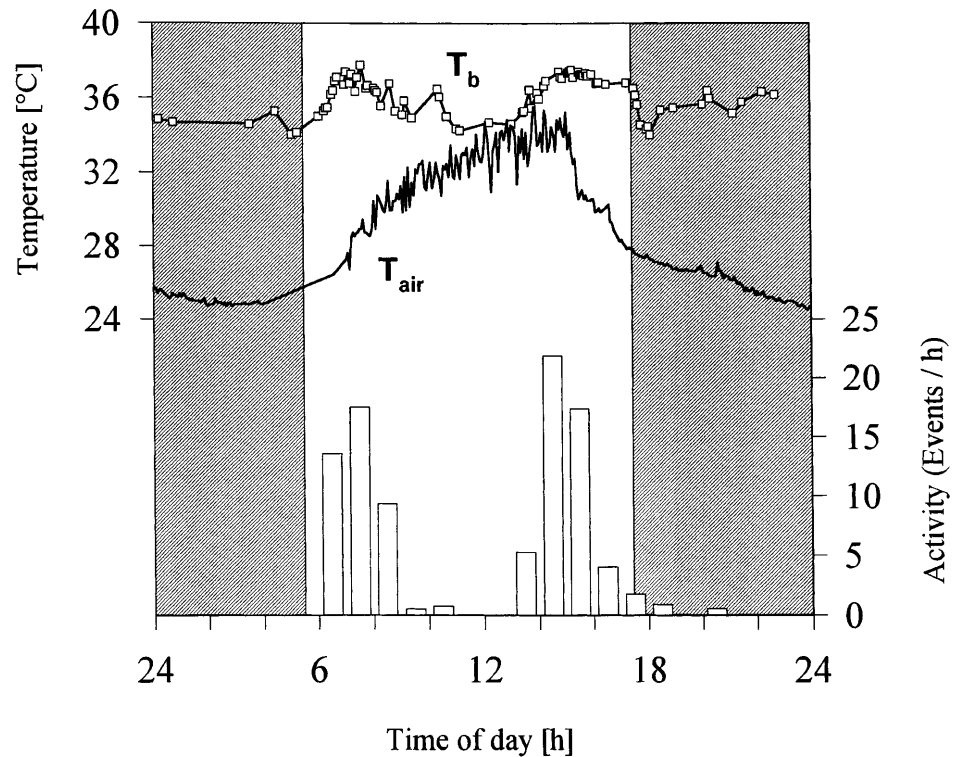


density of *A. cahirinus* was 20 individuals per hectare in July and October. Continuous monitoring of body weight showed no persistent effects of trapping activities or surgical treatment on *A. cahirinus* (Fig. 1). Mouse no. 0004 caught in two seasons (49.8 and 48.0 g, mean values in July and October, respectively) showed a slight increase in body weight of 4 g in July from trapping day

12 to day 23. Mean body weight of all trapped individuals was 33.9 g ($n = 26$).

A. cahirinus maintained a nocturnal T_b pattern throughout all seasons (Table 1, Fig. 5b; $P < 0.05$, Student's t -test). In January at night, T_b values reached a maximum of 38.2°C whereas a minimal T_b of 34.6°C was recorded during the photophase (no. 0004, Fig. 2).

Fig. 4 Example day of T_b , T_{air} and activity records for *A. russatus*, no. 0003 (day 8, Fig. 3, October 1996). Bars represent activity, expressed as number of positional changes detected with telemetry per hour



In July, T_b reached maximum values of 38.1°C during the nocturnal activity period (no. 0300). Between 0400 and 0600 hours, T_b decreased and varied from 34.0 to 35.2°C, although T_{air} in the sun rose to 44.6°C. Temperatures measured inside a rock crevice remained virtually constant at between 35 and 35.2°C throughout the entire day (see above). T_b increased again just after sunset. Individual T_b patterns showed no evident inter-individual differences in March ($n = 2$) but great individual variation in January ($n = 2$) and July ($n = 2$) (Fig. 5b).

Discussion

Individuals of *A. cahirinus* and *A. russatus* were retrapped frequently in their natural environment throughout all seasons of the year, indicating a rather sedentary behaviour which was not disturbed by trapping and surgery. Population estimation with the Jolly-Seber method indicates that almost total populations of both species were trapped and marked. Estimated population densities of *A. russatus* (38 individuals/ha during July, 35 individuals/ha during October) were higher than those of *A. cahirinus* (20 individuals/ha in July and October). This is consistent with previous studies (Shkolnik 1966; Shkolnik and Borut 1969; Kam and Degen 1993; Kronfeld et al. 1996) which showed abundances for *A. russatus* between 30–65 individuals/ha and *A. cahirinus* between 20–40 individuals/ha. These studies demonstrated that in areas where food is scarce, population densities of *A. russatus* are higher than those of

A. cahirinus. Previous measurements of mean body mass in the field varied for *A. cahirinus* between 33–35 g, for *A. russatus* between 42–44 g (Kronfeld et al. 1996; Shargal 1997). Mean body weights of all marked individuals in our research area are in agreement with these previous studies (*A. russatus* 48.2 g, *A. cahirinus* 33.9 g). In frequently retrapped individuals, we observed a slight increase of body mass in *A. russatus* shown in Fig. 1 during summer (9.7–27.4%) and fall (12.2–22.2%). This indicates a seasonality which is usually observed only in small mammals from moderate or northern latitudes (Heldmaier 1989).

We obtained continuous records of T_b and activity rhythms of spiny mice in their natural arid environment. Despite their close phylogenetic relationship, *A. cahirinus* and *A. russatus* differ with regard to their daily activity rhythms. *A. cahirinus* showed a clear nocturnal pattern with high T_b values during the hours of darkness. *A. russatus* showed elevated T_b s during daytime hours in the summer, and on some hot days a bimodal pattern could be observed.

Trapping observations in the field suggested that activity rhythms of both species are temporally segregated (Shkolnik 1971). When coexisting in the field, *A. cahirinus* is exclusively active during the dark phase, *A. russatus* during the light phase of the diel cycle. Body temperature measurements of *A. cahirinus* (Fig. 2) coincide with trapping results, and suggest that this species is nocturnal (Shkolnik 1971; Kronfeld et al. 1994). This is in agreement with other nocturnal desert rodents (Lovegrove et al. 1991), which are active during the night and thus do not experience heat load during the

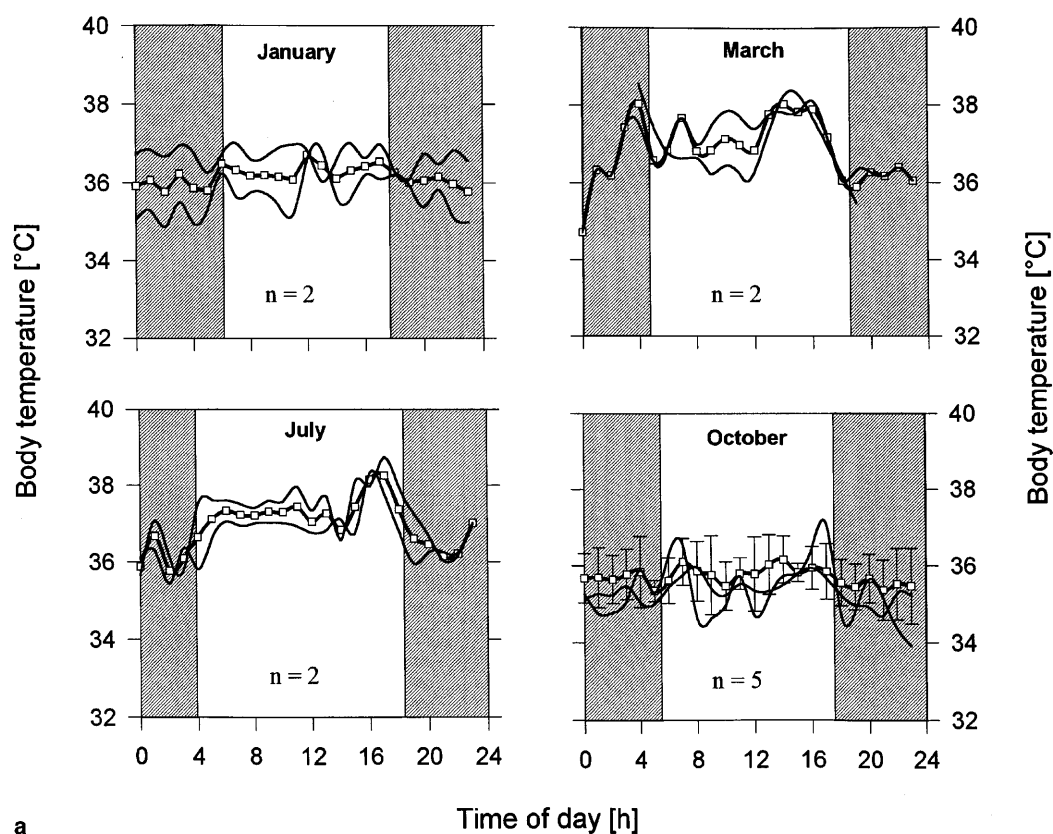
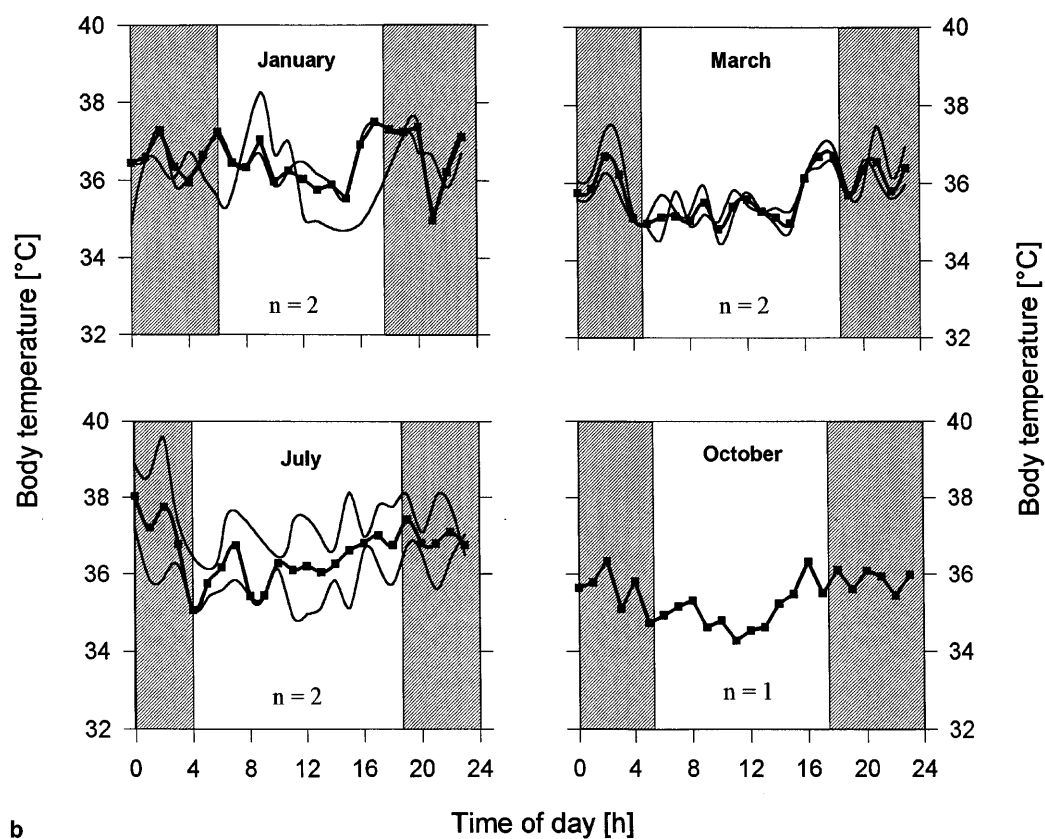
Acomys russatus*Acomys cahirinus*



Fig. 5 **a** Body temperature rhythms at all seasons from *A. russatus*. Data are shown as mean values per hour of all measured animals (open symbols). Mean T_b per hour of each measured individual is plotted additionally in each period except in October where $\pm$ SD is also shown. **b** Body temperature rhythms at all seasons from *A. cahirinus*. Data are shown as mean values per hour of all measured animals (closed symbols). Mean T_b per hour of each measured individual is plotted additionally in each period

hottest hours. *A. cahirinus* maintained its T_b at a low level during the hottest hours, indicating that the common spiny mouse was hiding in rock crevices during hot daytime hours.

A. russatus showed lowest levels of motor activity and core body temperature during the dark period and high values during daytime. This species displays a bimodal pattern of T_b and activity in the field, with maximum activity in the morning and late afternoon, and an extensive midday rest. T_b increased to a maximum of 41°C in July during daytime but decreased during the night to 33°C (Fig. 2). During the midday rest, T_b decreased to 36.4°C. This was close to the T_{air} measured about 1 m inside a rock crevice which showed rather constant levels of 35.0–35.5°C.

Most mammals, whether diurnal or nocturnal, exhibit activity peaks around dawn and dusk (Aschoff 1966). Two main activity patterns are described for diurnal rodents. *Psammomys obesus* manages to avoid extreme heat by being mainly crepuscular (Ilan and Yom-Tov 1990). Ground squirrels are active all day but reduce their activity when air temperatures are high (Degen 1996). T_b measurements run closely parallel to the states of wakefulness and sleep in diurnal and nocturnal species (Aschoff 1970; Honma and Hiroshige 1978; Gander and Moore-Ede 1983; Mc Elhinny et al. 1997). In small mammals, a close positive correlation between core body temperature and activity has been demonstrated. In rats, the circadian rhythm of T_b was characterised by bursts of small fluctuations which were well correlated with those of locomotor activity (Honma and Hiroshige 1978).

Large T_b fluctuations have been reported in active ground squirrels shuttling between sun and shade (Chappell and Bartholomew 1981). *Ammospermophilus*

leucurus typically shuttle between foraging and cooling sites. Foraging sites provide the squirrels with their energy sources, cool burrows or shade allow them to regain normothermia. A field study with *A. leucurus* even showed mean hourly T_b values of individual ground squirrels varying between 36.1 and 43.6°C in the summer (Chappell and Bartholomew 1981). After short activity periods aboveground, *A. leucurus* disappears from the surface to release the stored heat to the cooler environment in their shelters. They actually escape the heat and prefer cool microclimates. In nocturnal rodents like rats and golden hamsters (Gordon 1993; Refinetti 1995a, 1995b) and in a diurnal rodent, *Octodon degus* (Refinetti 1996), it was found that lower ambient temperatures are selected during the high phase of T_b rhythm. Behavioural observations showed that golden spiny mice moved very quickly from shade to shade (see Shkolnik 1971). When exposed to direct solar radiation, *A. russatus* experiences a very large heat load. Because of its small body size (mean body mass of 48.2 g) it has a low thermal inertia which would cause a rapid increase in T_b when aboveground during the hot part of the day (Degen 1996). *A. russatus* is therefore not a good candidate for shuttling behaviour as compared to larger-sized ground squirrels. The restriction of activity periods mainly to morning and afternoon hours prevents spiny mice from being exposed to severe heat load during midday hours.

Former studies have shown that *A. russatus* managed to maintain normothermia between 22 and 35°C T_{air} and survived 42°C T_{air} for at least 4 h (Shkolnik and Borut 1969). Laboratory studies showed that *A. cahirinus* tolerated temperatures between 5–30°C and became hyperthermic at T_{air} of 32.5°C. The lethal temperature was reached at 40°C T_{air} (Shkolnik and Borut 1969). Physiological responses measured under laboratory conditions have demonstrated that both species are well adapted to desert conditions, but some differences favour *A. russatus* over *A. cahirinus* for survival in hot deserts (Degen 1994). *A. cahirinus* is able to regulate its T_b over a wider range of T_{air} which explains the wide distribution, not only in deserts, but in high mountains where air temperatures fall below freezing at night (Shkolnik and Borut 1969). *A. russatus* can tolerate

Table 1 Mean T_b values for *Acomys* throughout the entire measuring periods in the photo- and scotophase

	Number of animals implanted with transmitters	Mean T_b (°C)	
		Day	Night
<i>A. russatus</i>			
March	2	37.54	36.48
July	2	37.32	36.48
October	5	35.79 $\pm$ 0.65 (SD)	35.66 $\pm$ 0.69 (SD)
January	2	36.25	36.08
<i>A. cahirinus</i>			
March	2	35.32	36.43
July	2	37.00	37.40
October	1	35.03	35.69
January	2	36.17	36.54

higher air temperatures than *A. cahirinus*. Both species have a lower than expected basal metabolic rate for rodents of their body mass [*A. russatus*: $0.73 \text{ ml O}_2 \text{ g}^{-1} \text{ h}^{-1}$ (Rubal et al. 1992); *A. cahirinus*: $1.1 \text{ ml O}_2 \text{ g}^{-1} \text{ h}^{-1}$ (Shkolnik and Borut 1969; Kleiber 1961)], but the extremely low field metabolic rate of *A. russatus* ($1.06 \text{ kJ g}^{-1} \text{ day}^{-1}$; see Degen 1994), compared to other rodents, matches its extreme desert distribution. Low metabolic rates reduce food and water requirements and internal heat production and prevent extensive hyperthermia. This may contribute to or even be a prerequisite for the diurnal activity and T_b pattern of *A. russatus*.

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